

OVERVIEW FOR TECHNICAL WRITERS

Common Control System (CCS)

One software suite to plan, fly, and manage the Navy's unmanned vehicles — and what that means for the people who document it.

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Notional overview drawn from public NAVAIR and trade-press sources. Not an official program product.

No CUI, classified, proprietary, or export-controlled information is contained in this document. Provided as a technical writing sample — Nathan B. Smith.

References

Each source and why it's here

NAVAIR — CCS product page

navair.navy.mil/product/CCS

Official mission statement, platforms (MQ-25, MQ-4C), open-architecture intent

NAVAIR — CCS first live demonstration

navair.navy.mil/node/23331

Evidence CCS spans domains: controlled a surrogate LDUUV; names PMA-281 and SSC (now NIWC) Pacific

C4ISRNET — Navy moves closer to CCS (2016)

c4isrnet.com

Plain-language split of vehicle management, mission management, mission planning

Forward Slope — NIWC Pacific SSA award

forwardslope.com

Shows NIWC Pacific's role as the CCS Software Support Activity (sustainment, releases)

AvWeek / FlightGlobal — MQ-8C

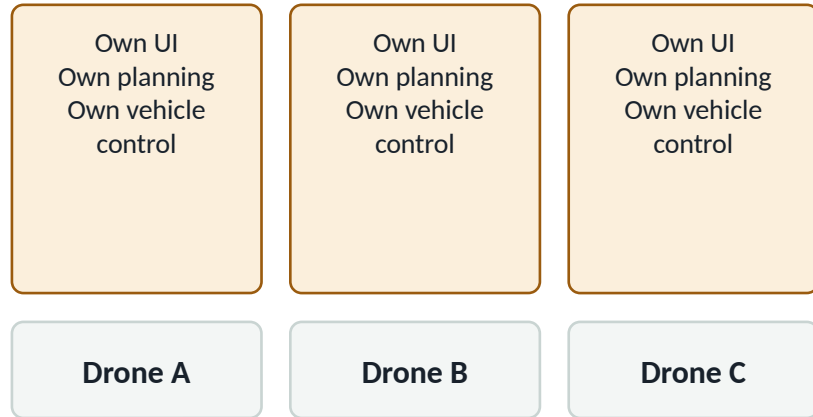
aviationweek.com · flightglobal.com

Corrects older sources: Fire Scout retires by end of FY2026

The problem CCS solves

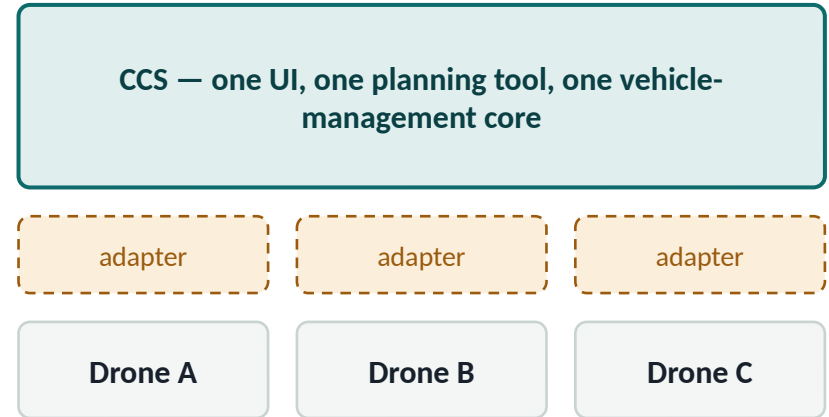
Before: one control stack per vehicle. After: one common stack, thin adapters.

BEFORE



Three manuals, three trainings, three sets of sailors.

AFTER



One core manual; only the platform-specific parts change.

Three core jobs

What the software does — and what operators will need from our documents

1

Mission Planning

Plan routes, tasks, and timing; re-plan as things change.

WRITERS DELIVER

Step-by-step planning procedures; worked examples

2

Vehicle Management

Fly or drive the vehicle; watch status and health.

WRITERS DELIVER

Control procedures, alerts and cautions, troubleshooting

3

Mission Management

Task the sensors; move the data — what to collect, when.

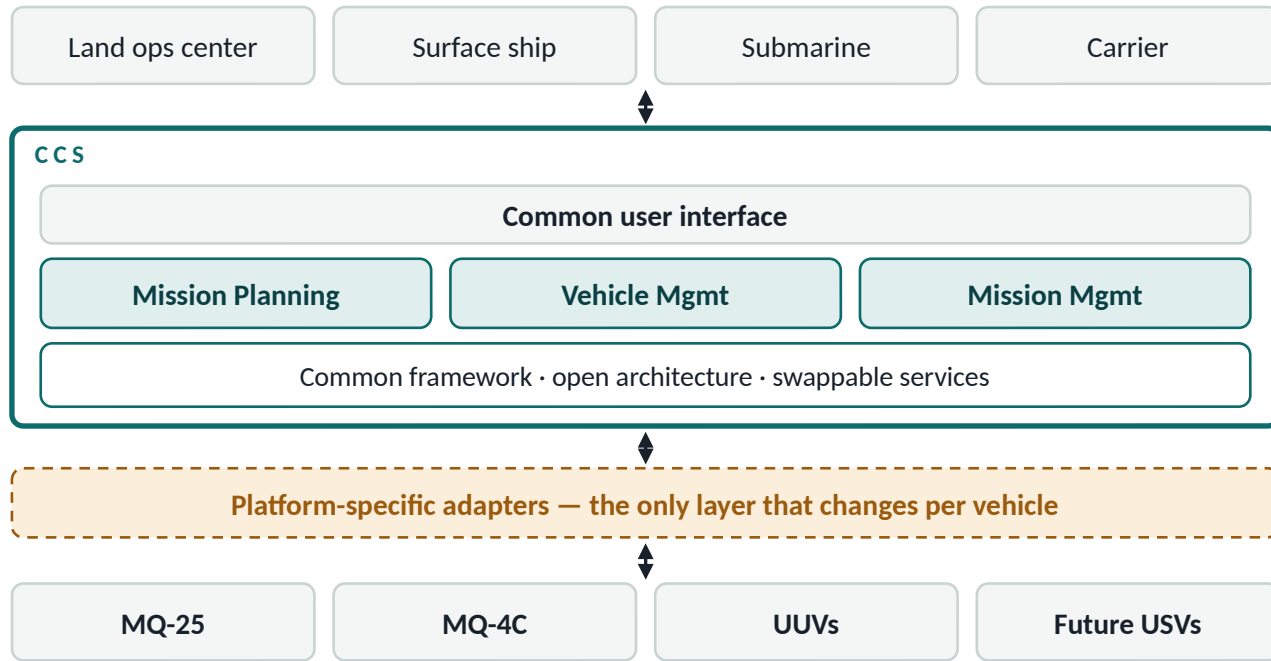
WRITERS DELIVER

Sensor tasking guides; data-flow explanations

How the pieces fit

Notional layers, top to bottom

CONTROL STATIONS



FOR WRITERS

Teal layers

= shared content. Write once, reuse everywhere.

Amber layer

= platform-specific content. Isolate it with conditions.

Stations

= different users and sites; same software.

Platforms and players

Who uses CCS, and who builds and sustains it

PLATFORMS

MQ-25 Stingray

Carrier-based unmanned tanker

MQ-4C Triton

High-altitude maritime surveillance

UUVs

Undersea; CCS demoed on a surrogate LDUUV

MQ-8C Fire Scout

Retiring by end of FY2026 — no longer a CCS growth path

PLAYERS

PMA-281

NAVAIR program office, Pax River — sets direction

NIWC Pacific

San Diego — sustains, releases, and fields the software

Contractor teams

Development, sustainment, CM, and documentation

Operators

Sailors in control stations — our readers

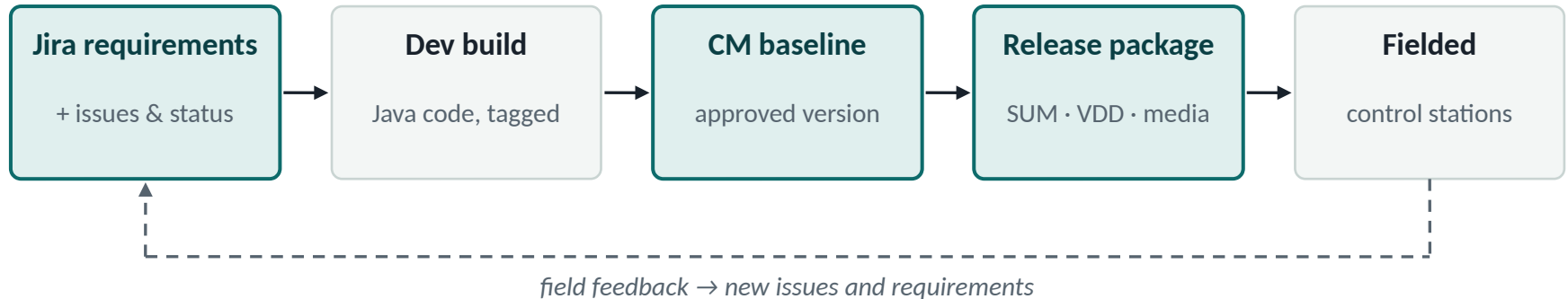
The documentation set

What a writer on CCS maintains — and what each one actually is

DOCUMENT	FORMAL NAME	WHAT IT ACTUALLY IS	READER
SUM	System/Software User Manual	How an operator uses the software, end to end.	Operators
How-To Guide	Task-focused procedures	One job, start to finish: “plan a route,” “task a sensor.”	Operators
Training docs	Courseware and lesson material	Teaches the tasks before operators do them for real.	Trainees, instructors
VDD	Version Description Document	The “what’s in this box” for each release: version, changes, known issues.	Sites, CM, sustainment
Install docs	Installation & configuration	How to load the release at a site and set it up.	Site admins
Status reports	From the Jira requirements DB	Requirements, issues, status — for decisions on what gets fixed next.	Program leadership

Where writers sit: the release loop

Writers work on both sides of the build



Teal = the writer's steps. Going in, keep requirements and issues accurate so reports are trustworthy. Coming out, make sure the SUM, VDD, and media label all carry the same version as the CM baseline.

Tools and constraints

How the architecture turns into a single-source authoring setup

Structured authoring (DITA)

Topic-based source on the program network; no cloud services under CUI

Jira + Confluence

Requirements, issues, status reports, working drafts

Git / CM system

Doc project under the same baseline and tags as the code

Classified media

Package, mark, and transmit media per site SOP

SINGLE-SOURCE FEATURE → CCS USE

Filtering	MQ-25 vs. Triton content in one project
Keys	Release version in one place
Content reuse	Shared warnings, common UI steps
Output templates	CUI banners and designation block
Build targets	SUM as PDF, How-Tos as offline HTML5

Key takeaways

1

Common core, thin edges.

Most CCS content is shared; only the adapter layer is platform-specific.

2

Architecture = content strategy.

Shared layers become reusable topics; platforms become filtering attributes.

3

Versions must match.

Code tag, CM baseline, SUM, VDD, and media label all agree.

4

Write for operators.

Sailors in control stations are the reader; procedures beat descriptions.